



Feed the Future Innovation Lab for Livestock Systems

Exposure Assessment of Campylobacter Infections in Rural Ethiopia (EXCAM) Project

Standard Operating Procedures (SOP) for Biological Human Sample Collection

SOP # E3.3A

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Experimental Location(s): Ethiopia

SOP Purpose

The purpose of this document is to describe the technical procedures required to collect representative human-associated samples from the selected cohorts during the "Campylobacter Genomics and Environmental Enteric Disfunction" (CAGED) longitudinal study and "Exposure Assessment of *Campylobacter* Infections in Rural Ethiopia" (EXCAM) cross sectional study. This SOP will outline the different methods of sample collection based on the type of sample being collected for sample processing, provide sample collectors with information on equipment, materials and data sheets needed for sample collection and personal protective equipment (PPE) requirements.

Risk Assessment and Safety Measures

Each SOP should include a detailed outline of the potential safety hazards to ensure the safety of all team members, participants, and samples by mitigating the listed potential safety hazards. An example of safety procedures should include using proper personal protective equipment (PPE) such as gloves, shoe covers, or a face mask.

RISK ASSESSMENT:

EXPOSURE FACTORS	ENVIRONMENT FACTORS	ANIMAL/PLANT FACTORS	PROCESS RISKS
☐ Oral transmission route	☐ Environmental stability	☐ Amplifying host	☐ Cryogen
☐ Percutaneous transmission route	☐ Reproducible agent/vector	☐ Agent shedding	☐ Corrosives
☐ Mucosal transmission route	☐ Endemic host/vector	☐ Physical hazards (teeth, claws, etc.)	☐ Flammables
☐ Inhalation transmission route	☐ Drug-resistant pathogen	☐ Exotic organism	☐ Oxidizers
☐ Person-to-person communicable	☒ Human pathogen	☐ Noxious/invasive organism	☐ Carcinogens
☐ Fomite	☐ Animal pathogen	☐ Field study	☐ Radionucleotides
	☐ Plant pathogen		

PERSONAL PROTECTIVE EQUIPMENT:

☐ Re-usable lab coat/gown	☐ Surgical mask
☐ Disposable lab coat/gown/coverall	☐ N95 respirator

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☒ Disposable (latex/nitrile) gloves	☐ Powered Air Purifying Respirator (PAPR)
☒ Safety glasses/goggles	☐ Heat-resistant gloves
☐ Face shield	☐ Cold-resistant gloves
☐ Hearing protection	☒ Other: Hand Sanitizer
ENGINEERING CONTROLS:	
☐ Biological safety cabinet (BSC)	☐ Aerosolization chamber
☐ Fume hood	☐ Autoclave
☐ Cages/Isolators/Growth chambers	☐ Hand-washing sink
☐ Facility directional airflow	☐ Separation of rooms/ functions (i.e. for PCR work)
	☒ Other: Hand Sanitizer
Definitions, Abbreviations and Acronyms The term drag swab is used repeatably to describe the sampling kit used for areola sampling.	
Responsibilities All designated team members are required to read the SOP in detail before executing the specified procedure. Once the team member has read the SOP at length, they will be required to sign the SOP verification sheet under records management (section 8) within the SOP. The project manager or team leader will verify everyone's knowledge and training of the SOP before procedure execution. It is the responsibility of both the staff, project manager and team leaders to ensure that the procedure or process follows the details noted in the individual SOP and to detail in writing when the SOP or a component of that SOP has not been followed along with reasoning for SOP deviation. <i>Additional Responsibilities</i> - All sample collectors performing sample collection practices must adhere to the safety practices and PPE listed above. - All sample collectors must fill out all sample collection logs at the site of collection.	
References There are currently no references available to support this document.	

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Procedure

EXCAM will enroll a total of 78 participants (13 participants per cohort). EXCAM will sample cohorts for five human-associated samples (hand wash/rinse for child/sibling/ and mother, breast swab and breast milk) will be collected randomly two times within the study timeline. These samples will be collected once at when the child is 1-3 months of age and again at 6-9 months of age.

• Record Equipment

Item	Location	Special Notes
Cooler boxes, Ice packs, and Water	Sample Reception Room	Used for sample transportation
Foldable table	Sample Reception Room	Workspace for sample collection (if needed)

• Record Materials, Supplies, and Reagents

Item	Information/Location	Purpose (details can be found in section E)
Whirl pak-bags	VWR Catalog#66174-612	Hand rinse/wash sample collection, breast swabbing collection
Drag swabs	VWR Catalog # 89221-704	Breast swab sample collection
Specimen Containers	VWR Catalog# 10811-012	Breast milk sample collection
Saline	Prepared in the lab	Hand rinse/ wash and Breast swab sample collection
Gloves	VWR Catalog # 89235-580	PPE
Hand Sanitizer		Hand washing
70% Ethanol		Disinfecting work surfaces
Paper Towels		Disinfecting work surfaces and Breastmilk sampling
Alcohol Wipes		Disinfecting surrounding areas for breastmilk sample collection

• Preparation

Sample collection (n=5; Hand wash, breast swab, and breast milk) will require the use of a variety of materials that will be prepared in advance by lab personnel (**Table 2**). Sample collectors will collect prepared materials for sample collection either the day of

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sample collection or the day before and verify with **Appendix 1. Sample Collection Materials Checklist** before starting field work.

Table 2. Tools required for the collection of the human-associated samples

Sample type	Preparation	Storage condition during transport
Hand Rinse	Whirl pack bag prefilled with 200mL of saline	Ice cooler containing water and ice packs
Areola Swab	Drag swab that contains 4mL of Letheen Broth, swab and tube	Ice cooler containing water and ice packs
Breastmilk	Sterile specimen container	Ice cooler containing water and ice packs

Laboratory personnel will prepare the sample collection materials in advance to ensure that the collection tubes and bags have proper reagents for sample transportation. All prepared sample collection materials will be recorded in **Appendix 1. Sample Collection Materials Checklist**. Materials will be subdivided into designated project bins to assist with organization during field work. All materials will be labeled by lab personnel with the name of reagent/medium and color coded (Green for EXCAM).

All human biological samples collected for the purposes of EXCAM, must be labeled with sample ID, date of collection, type of sample, and project. This information must record in to track collected samples. Please review **Appendix 2** for more details on **Human Biological Sample Collection Log**.

Figure 1. EXCAM Biological Human Sample Label

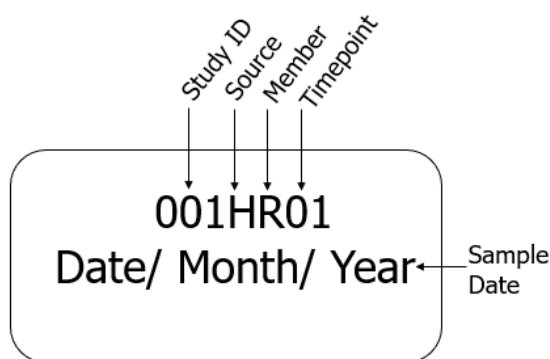


Table 3. EXCAM Sample Labeling Summary

Sample Type	Member ID	Sample Description	Project
Human Samples	R	Hand Rinse (Child) Sampling	EXCAM
	H	Hand Rinse (Sibling) Sampling	EXCAM
	P	Hand Rinse (Mother) Sampling	EXCAM
	A	Areola Swab Sampling	EXCAM
	B	Breastmilk Sampling	EXCAM

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Figure 1. Represents an example of the human sample labeling scheme that will be utilized by EXCAM. The Study ID will be based on participant enrollment and the Source ID (H) represents it is a human sample.

Labeling for Biological Human Sampling

All tubes should be labeled prior to sample collection in the field. Verify that the label matches the sample type and ID as indicated by the project manager prior to sampling. This information must be recorded, to track collected samples. Please review **Appendix 2** for more details.

- **Routine Disinfection in the field**

The field team must wear appropriate PPE (gloves, lab jacket, and shoe covers; listed in section 3) and set up the workspace before collecting samples in the designated household. Disinfection of the field team (hand and boots) and tools must be performed before and after collecting a sample in the designated household. Similarly, the field team and tools must be disinfected once arriving and before leaving a household.

The disinfection of the field team and tools must be performed using 70% ethanol and paper towels. Allow the disinfected area/tools to dry before starting the sampling procedure.

- **Collection of the sample in the field**

Below you will find the detailed methods for sample collection associated with each specific type of sample.

Hand wash/rinse sampling (please review hand wash sampling video and Appendix 3 for more details):

Infant Hand washing

- Check that the sample bag label is clearly visible.
- Unzip the sampling bag and insert the infant's hands into the sample bag without touching the solution inside (200mL of sterile water).
- From the outside of the bag, softly squeeze the bag around the infant's hands to mimic washing their hands for at least 30 seconds using the solution.
- Check that all the parts of the hand, including fingers, palm and fingernails are fully immersed in the solution and well.
- Once complete, remove the hands from the sample bag and carefully close the sample bag.

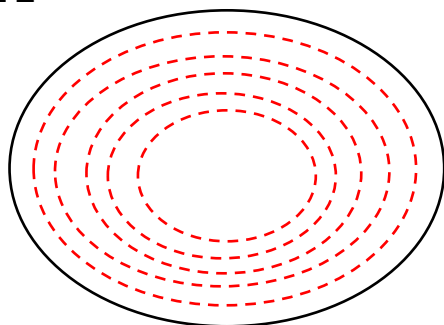
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• Transfer the contents of the large sampling bag into a medium whirl Pak bag by pouring the liquid from the big bag to the smaller bag. Carefully roll the sample bag and secure by using the yellow tabs on the sides. • Store the sample bag in the icebox with an ice pack and water during the field sampling. • Wash the child's hands with water and dry using paper towels or use wipes. • Record samples in Appendix 2. Human Biological Sample Collection Log. Sibling and Mother Hand washing • Check that the sample bag label is clearly visible. • Unzip the sampling bag and ask the sibling or mother to insert their hands in the sample bag and mimic washing their hands for at least 30 seconds using the solution. • Carefully check to insure all the parts of the hand, including fingers, palm and fingernails are fully immersed in the solution and well. • Once complete, assist sibling and mother with removing their hands from the sample bag. • Transfer the contents of the large sampling bag into a medium whirl Pak bag by pouring the liquid from the big bag to the smaller bag. Carefully roll the sample bag and secure by using the yellow tabs on the sides. • Store the sample bag in the icebox with an ice pack and water during the field sampling. • Wash the sibling and mothers' hands with water or wipes and dry using paper towels. • Record samples in Appendix 2. Human Biological Sample Collection Log. <i>Areola swabbing (see Appendix 4 for a step-by-step guide with pictures):</i> *This can be done by the mother under observation by the sample collector. If the mother prefers to collect the sample independently, the sample collector must discuss the following steps and observe that the mother has cleaned the area and hands before collecting the sample. * • Check that the drag swab label is clearly visible. • Carefully twist the drag swab to remove the swab from the container. • Perform the areola swab by rotating the swab in a circular pattern (on both areola). Start the areola swab from the outside extremity of the areola and going towards the middle of the areola (the nipple; Figure 2). • Avoid touching the swab to the surrounding skin on the breast.
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- Place the swab in the sample container. Twist to seal.
- Store the sample bag in the icebox with an ice pack and water during the field sampling.
- **Record samples in Appendix 2. Human Biological Sample Collection Log.**
- **This is to be done on both areolas using the same swab.**

FIGURE 2



Follow the direction of the red lines for areola swab sampling.

Breast milk collection (see Appendix 5 for a step-by-step guide with pictures):

***This can be done by the mother under observation by the sample collector. If the mother prefers to collect the sample independently, the sample collector must discuss the following steps and observe that the mother has cleaned the area and hands before collecting the sample. ***

- Check that the sterile specimen container is clearly labeled.
- Place a paper towel on the mother's lap.
- The mother must wear gloves and sanitize them with alcohol before starting the breast sampling.
- Clean the areola with an alcohol wipe (allow the alcohol to dry).
- The mother must re-sanitize the gloves with alcohol before starting the breast sampling.
- Position the dominant hand in a "C" shape, with the pads of the thumb and fingers about 1.5 inches (3.8 cm) behind the areola (**Figure 3**).
- Push straight back into the chest wall (**Figure 4**).
- Roll thumb and fingers forward to milk the breast (**Figure 4**).
- Allow the first few drops to fall/discard onto the towel (**Figure 4**).
- Collect "mid-stream milk" (**as such as possible**) into a sterile specimen container (**Figure 4**). Ensure that breastmilk is collected from both breasts (a combined total of 10mL of sample).

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- Verify that the label is still legible and close the sample tube.
- Store the sample in the icebox with an ice pack and water during the field sampling.
- **Record samples in Appendix 2. Human Biological Sample Collection Log.**

Figure 3. Hand positioning for manual breast milking.

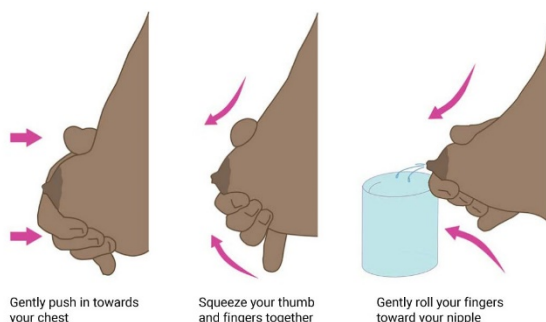


Figure 4. How to squeeze milk from breast (note: the second hand should hold breast steady but is not in this picture).

• **Length of Procedure**

The length of the procedures will vary depending on the participants. On average, they will likely last for the length of time below:

Type of Sample	Method	Time
Hands (Child/Sibling/Mother)	Hand rinse/Wash	5-10 minutes
Areola	Areola swab	5-10 minutes
Breastmilk	Manual squeezing	5-10 minutes

Overall, between explaining the process to participants, preparing samples, collection, and storage, human sampling should take no more than 1 hour per household.

• **Sample Transportation**

All samples collected in the field will be stored in a cooler box that contains ice packs and water during field work and transportation to the lab for processing.

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• Receiving of the samples at HU All samples will be received in the lab by lab personnel and recorded using the sample logbook. Field collection recordings will be scanned and saved electronically.
Emergency Procedures In case of emergencies during field collection please contact the project manager immediately at phone number +251 91211 5740. All emergencies must be documented and reported to the principal and co-investigators in a timely manner.
Records Management A. SOP LOCATION: A hard copy of the SOP will be kept in the sample collection log binder to be easily accessed in the field. An additional hard copy will be kept in the lab for reference. B. TRAINING/ SOP VERIFICATION SHEET: Each team member shall receive training at the time of initial assignment to the lab and prior to assignments involving new procedures. This training shall include, but not be limited to the hazards identified in the Risk Assessment, extensive review of the SOP, measures team members must follow to protect themselves from exposure to these hazards, and availability of reference materials, such as Material Safety Data Sheets. All training and review of SOP by the team member will be recorded in the SOP Distribution List The SOP Distribution List must be completed and saved as a record of the training and review of SOP. If the Project Manager/ Team Leader/ or PI/ CoPI feel that the team member requires an SOP refresher the team member will need to complete the SOP Verification form again to document the refresher training.
Revision History/ Rationale No revision currently available.

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Appendix 1. Sample Collection Materials Checklist

Material Preparation Date: _____ Sampling Date: _____

Sampling Locations: _____

Additional Remarks:

Sample Collection Materials Checklist				
Materials	Quantity	Prepared by (Lab Initials)	S. Collector Initials	Remarks
Zipper plastic bags				
Whirl pak-bags				
Drag Swabs				
Specimen Container				
Peptone water				
Gloves				
Hand Sanitizer				
70% Ethanol				
Paper Towels				
Alcohol Wipes				

Appendix 2. Human Biological Sample Collection Log

Sampling Date: _____ Sampling Locations: _____

Sample Collector(s): _____ Lab Sample Arrival Time: _____

EXCAM Human Sampling									
No	Sample ID	Human Sample Collection Type					Sampling Time	S.C. Initials	Comments
		Hands R	Hands H	Hands P	Areola \ swab	Breastmilk			
1									
2									
3									
4									
5									

Please note that all sample collectors must record the following information:

No. represent the number of sample vials used.

Household ID are assigned by the project manager.

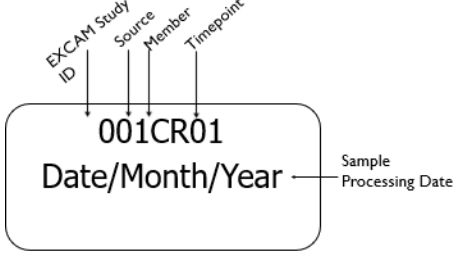
Sampling time represents the time in which the samples were collected.

S.C Initials help to identify the sample collector.

Comments Represents any comments that need to be documented during sample collection such as animal owner was not present during sampling time, collected animal samples came from neighbors, etc.

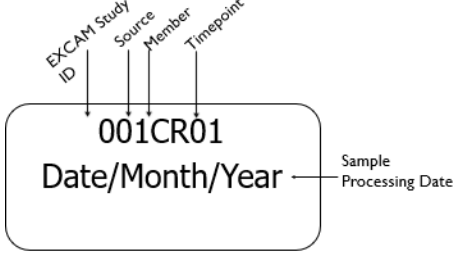
Appendix 3. Hand Wash/Rinse Methods

Infant

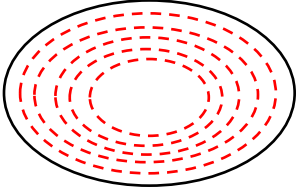
	Check both sample bag labels (large bag with white tabs, small bag with yellow tabs). Be sure labels clearly visible.
	Unzip the sampling bag and insert the infant's hands into the sample bag without touching the solution inside.
	Insert the infant's hands without touching the solution inside
	Squeeze the outside of the bag around infant's hands for at least 30s • Check that all parts of the hands are submerged in the solution (hand, fingers, palm, fingernails)
	Remove hands from sample bag, transfer contented to small sample bag with yellow tabs. Close bag and secure with yellow tabs.
	Store the sample bag in the icebox with icepack and water
	Rinse infant's hands with sterile water
	Record samples in Appendix 2. Human Biological Sample Collection Log.

Appendix 3. Hand Wash/Rinse Methods (continued)

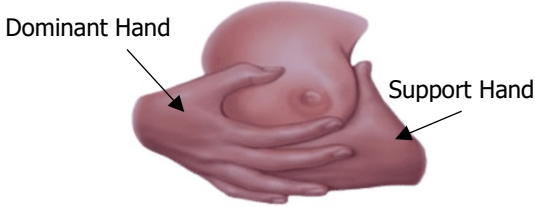
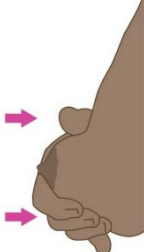
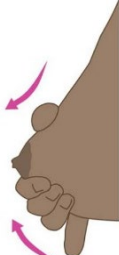
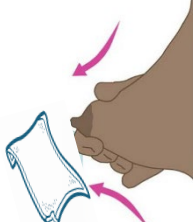
Sibling and Mother

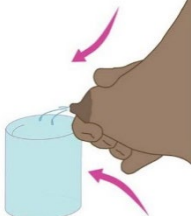
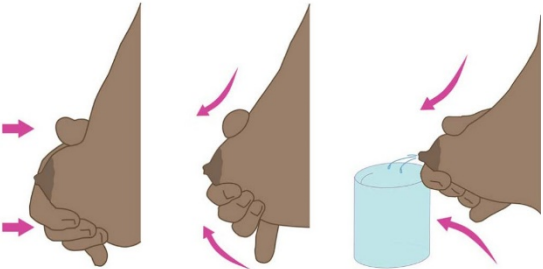
	Check both sample bag labels (large bag with white tabs, small bag with yellow tabs). Be sure labels clearly visible.
	Unzip the sampling bag and insert the infant's hands into the sample bag without touching the solution inside.
	Have sibling or mother put their hands in the bag and mimic washing their hands for 30s • Check that all parts of the hands are submerged in the solution (hand, fingers, palm, fingernails)
	Assist them in removing hands from sample bag, transfer contented to small sample bag with yellow tabs. Close bag and secure with yellow tabs.
	Store the sample bag in the icebox with icepack and water
	Rinse participant's hands with sterile water
	Record samples in Appendix 2. Human Biological Sample Collection Log.

Appendix 4. Areola Swab

	Check that the drag swab label is clearly visible.
	Carefully twist the drag swab to remove the swab from the container.
	Perform a ten-point culture of the areola. DO NOT CLEAN THE AREOLA. Rotate the swab in a circular pattern . Repeat on second areola with same swab. Avoid touching swab to skin outside the areola
	Place the swab in the sample container. Twist to seal.
	Store the sample bag in the icebox with an ice pack and water during the field sampling.
	Record samples in Appendix 2. Human Biological Sample Collection Log.

Appendix 5. Breastmilk Sample

	Check that the sterile specimen container is clearly labeled
	Place paper towel in mother's lap
	Clean areola and breast with sterile saline and dry with sterile gauze
	Clean each areola with an alcohol wipe. Allow to dry.
	Position the dominant hand in a "C" shape with the pads of the thumb and fingers about 3.8cm (1.5in) behind areola. The second hand should support the breast.
	Push that hand straight back into chest
	Roll thumb and fingers forward while squeezing
	Let first few drops fall on towel

	Collect ~5mL of milk into the specimen container
 Gently push in towards your chest Squeeze your thumb and fingers together Gently roll your fingers toward your nipple	Repeat on second breast with the same container. Collect around 10mL milk total
	Verify the label is still legible. Close tube.
	Store sample in the icebox with an ice pack and water during the field sampling
	Record samples in Appendix 2. Human Biological Sample Collection Log